

STATES AND GOALS

No.	Room State	Sensor State	Goal
1	Normal	Regular temperature and humidity, air quality, and patient vitals indicate a normal room state.	Determine the goal based on the patient 's state.
2	Hazardous	Either the air-quality or temperature goes outside safe levels indicating gas leak or fire hazard in the room, and the ambience unsafe.	Raise high-level alarm. Unlock the door for easy access. Set the light brightness to max.
3	Emergency	The patient activates a manual emergency by pressing the SOS button. This indicates that the patient requires urgent assistance.	Raise high-level alarm. Unlock the door for easy access. Set the light brightness to max.

No.	Patient State	Sensor State	Goal
1	Awake	Pressure Sensor detects the patient is in bed, and the PIR sensor detects regular movement every few minutes detecting awake state.	Keep the patient in a well-lit room and at comfortable temperature.
2	Resting	Pressure sensor detects the patient is in bed, and the PIR sensor detects no movement for over 15 minutes, indicating resting state.	Turn off the lights and keep the room well-ventilated and focus on saving power.
3	Out-of-bed	Pressure sensor detects no patient in bed. The heart monitor detects regular vitals and heart rate indicating the patient out of the bed.	Wait for 15 minutes, if the patient is not back in bed, raise a low-level alarm.
4	Distress	Irregular heart rhythm or low oxygen level indicates the patient in distress.	Raise high-level alarm to alert the nurse and on-call doctor. Unlock the door for easy access.

ACTIONS FOR AI PLANNING

EVERY TIME THE STATE OF ROOM/PATIENT CHANGES, RE-PLANNING IS TRIGGERED. PLANNER OUTPUT CAN DO ONE OR MORE OF THE FOLLOWING ACTIONS:

1. SET LIGHT BRIGHTNESS **X** (0% - 100%)
2. SET FAN SPEED **Y** (0% - 100%)
3. SET DOOR STATE - (**LOCK** / **UNLOCK**)
4. SET BUZZER STATE - (**OFF**, **Low-level alarm**, **High-level alarm**)

ACTUATOR IN DIFFERENT STATES

ACTUATOR	STATE	FACTORS/VALUE
LIGHT	Room - Hazardous	Set brightness to 100% by AI planner
	Room - Emergency	Set brightness to 100% by AI planner
	Patient - Distress	Set brightness to 100% by AI planner
	Patient - out-of-bed	Not affected in this state* (<i>*No AI-Planner intervention</i>)
	Patient - Resting	Set brightness to 0% by AI planner
	Patient - Awake	If set to manual mode, set brightness determined by user If set to auto mode, use the light sensor to get sunlight level and determine the brightness accordingly using AI planner to ensure sufficient lighting with minimum possible power consumption.
FAN	Room - Hazardous	Not affected in this state*
	Room - Emergency	Not affected in this state*
	Patient - Distress	Not affected in this state*
	Patient - out-of-bed	Not affected in this state*
	Patient - Resting	If set to manual mode, set fan speed determined by user If set to auto mode, use the DHT sensor to get temperature and humidity and use AI planner to determine fan speed for sufficient ventilation with minimum possible power consumption.
	Patient - Awake	If set to manual mode, set fan speed determined by user If set to auto mode, use the DHT sensor to get temperature and humidity and use AI planner to determine fan speed for comfortable ambient temperature.
DOOR	Room - Hazardous	UNLOCKED by AI planner
	Room - Emergency	UNLOCKED by AI planner
	Patient - Distress	UNLOCKED by AI planner
	Patient - out-of-bed	Not affected in this state*
	Patient - Resting	LOCKED by AI planner it unless authorized person (doctor/nurse/family visitor) unlocks it manually from outside
	Patient - Awake	Not affected in this state, LOCKED/UNLOCKED without AI planner by patient, authorized person (doctor/nurse/family visitor) unlocks it manually from outside

BUZZER	Room - Hazardous	High-level alarm by AI planner
	Room - Emergency	High-level alarm by AI planner
	Patient - Distress	High-level alarm by AI planner
	Patient - out-of-bed	Low-level alarm by AI planner if this state persists for more than 15 minutes
	Patient - Resting	OFF by AI planner
	Patient - Awake	OFF by AI planner